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Next-Gen Capability in Well Construction: A Framework for Future Skills and Reskilling

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Abstract

The evolving complexity of well construction demands a new approach to reskilling and upskilling technical professionals. As the industry advances through digitalization and operational integration, sustained performance depends not solely on tool proficiency, but on the ability to work across disciplines, interpret complex data environments, and influence outcomes through collaborative and adaptive leadership. A modular framework is proposed to develop integrated capability across three domains: technical fluency, cross-functional integration, and behavioral effectiveness.

The framework is informed by a longitudinal review of skill evolution over the last decades, combined with field observations and structured interviews across drilling, subsurface, and digital teams. A capability mapping exercise and performance-linked analysis were used to define priority skill areas. The resulting model features a modular, tiered structure designed to support learning across career stages, with mechanisms for performance feedback and regional adaptation.

Rather than treating digital systems, technical knowledge, and behavioral skills as separate tracks, the framework positions digital fluency as an enabler that enhances engineering judgment, accelerates team coordination, and strengthens decision quality. This integrated view addresses common inefficiencies in well delivery—such as repeated design iterations, slow response times, and fragmented execution—by equipping professionals to operate in high-complexity environments with confidence and agility. The proposed model is both scalable and adaptable, aligning workforce development with the operational realities of next-generation well construction.

Introduction

The well construction domain is undergoing a fundamental shift—driven not only by technological innovation, but by the increasing integration of disciplines, compressed delivery timelines, and the expanding scope of decision-making at the front lines. Technical professionals are now expected to operate in multidimensional environments where outcomes depend on the ability to synthesize input across geology, engineering, completions, and operations.

Traditional models of professional development - grounded in field exposure, domain specialization, and progressive technical responsibility - are increasingly insufficient for the demands of modern well delivery.

While structured training remains common, many programs focus heavily on digital tools, procedural updates, or regulatory compliance, offering limited emphasis on broader integrative or behavioral capabilities.

Crucially, digital systems are no longer peripheral - they shape the information flow, influence decisions, and define interfaces across technical teams. However, tool proficiency alone does not ensure effectiveness. What differentiates high-performing professionals is the capacity to interpret data in context, challenge automated output where needed, and align technical intent with field realities.

To address these evolving demands, a structured and integrated capability model is introduced. It integrates three core dimensions: technical fluency, cross-functional integration, and behavioral leadership. Digital fluency is treated not as an isolated track, but as an embedded enabler that enhances judgment, accelerates coordination, and sustains performance in increasingly complex operational settings.

Historical evolution of skills

The progression of skill requirements in well construction reflects broader shifts in operational priorities, technology adoption, and system complexity. Over the past five decades, the expectations placed on technical professionals have expanded from execution to integration, and from task-specific delivery to system-level contribution.

In the 1970s through the early 1990s, engineering roles were centered around mechanical execution and operational responsiveness. Engineers acquired practical expertise through direct field exposure, informal mentoring, and hands-on problem-solving. The boundaries between office and field were minimal, reinforcing alignment between design and execution. The prevailing model emphasized the engineer as a technical practitioner operating within a single domain of responsibility.

From the late 1990s to the 2010s, the emphasis shifted toward planning, standardization, and assurance. The growing use of simulators, digital planning tools, and procedural systems moved engineers into more structured roles, with greater involvement in documentation, assurance workflows, and front-end design. Technical competency frameworks began to formalize, but engagement across functions remained limited.

As the industry entered the 2020s, technical environments became more interdependent and data-rich. Disciplines such as drilling, subsurface, and completions increasingly overlapped in both design and execution. The role of the engineer began to evolve beyond domain-specific tasks into cross-functional coordination and decision integration—requiring not only subject matter expertise, but also systems thinking and adaptive collaboration.

While foundational technical knowledge remains a cornerstone of professional development, it now operates within a broader competency landscape shaped by interconnected disciplines, real-time data environments, and evolving operational demands. Professionals are increasingly expected to interpret complex information streams, collaborate across functional boundaries, and guide decisions under conditions of uncertainty. In this context, digital systems serve as accelerators - enhancing the speed and volume of information—but it is human capability that remains central to effective interpretation and application. The ability to integrate knowledge across domains, apply systems thinking, and exhibit behavioral adaptability is emerging as a defining characteristic of high-performance well delivery professionals.

Identifying the Gaps in Current Reskilling Approaches

While many organizations have invested in workforce development, existing reskilling approaches may not fully address the demands of modern well delivery. Numerous programs offer technical refreshers, software training, or e-learning modules; however, they frequently remain fragmented in scope - focusing on tool usage, procedural updates, or regulatory compliance rather than on broader professional development.

One underlying limitation is the target audience and delivery model of most training programs. These initiatives are typically structured for early-career professionals and new graduates, providing them with foundational knowledge and onboarding support. In contrast, more experienced personnel often receive selective or project-specific training, driven by immediate operational requirements rather than a structured, long-term development plan. This creates inconsistency across the organization and limits the broader transfer of knowledge and capability.

A further gap lies in the limited emphasis on cross-functional collaboration and integrated decision-making. As well construction becomes increasingly interdependent with subsurface, completions, and production planning, engineers must be equipped to understand the broader impact of their design and operational choices. However, most training programs continue to operate within disciplinary boundaries, offering few opportunities for multidisciplinary engagement or collaborative problem-solving in realistic scenarios.

The behavioral dimension of technical work also remains underrepresented. While modern delivery environments demand strong communication, adaptability, and influence across teams, these competencies are rarely treated as integral to technical excellence. When included, behavioral elements are often delivered as standalone workshops or optional learning modules, rather than embedded within the technical learning journey.

Finally, there is often a disconnect between training and operational implementation. Many programs do not incorporate field-based feedback loops or link learning outcomes to measurable business performance. As a result, participants may complete training without a clear pathway to apply new knowledge or demonstrate impact - undermining both motivation and organizational value.

Together, these gaps highlight the need for a more holistic, consistent, and performance-linked approach to capability development - one that integrates technical, behavioral, and digital growth across all career stages.

Defining the Three Core Skill Domains

Building a high-performance technical workforce for modern well construction requires a shift from fragmented or tool-centric training toward a more integrated capability model. Based on field observations, stakeholder input, and analysis of evolving operational demands, three core domains of competence emerge as critical for the next generation of professionals: technical fluency, cross-functional integration, and behavioral agility and decision influence. These domains are not independent silos—they reinforce each other and are increasingly interwoven through the use of digital systems that shape how decisions are made, data is shared, and teams operate.

Technical Fluency

Technical fluency remains the foundation of engineering practice. It includes deep understanding of well design principles, drilling mechanics, fluids, pressure management, and geomechanics. However, it now also requires the ability to work effectively within digital planning environments, validate automated design outputs, and apply domain knowledge to rapidly assess data trends.

Modern engineers must move beyond calculation-based understanding and develop diagnostic thinking: interpreting system behaviors, anticipating risks, and adjusting designs based on evolving conditions. This domain emphasizes both analytical rigor and operational context - and increasingly relies on the ability to interface with digital platforms that support modeling, design iteration, and decision framing.

Cross-Functional Integration

Well delivery is no longer a sequence of isolated technical steps - it is a system of interdependent functions. Drilling outcomes are influenced by geomechanical constraints, reservoir priorities, completions strategies,

and even surface facility considerations. Cross-functional integration refers to the technical professional's ability to understand these interfaces and contributes to collective optimization.

This competency area includes the ability to engage in collaborative design reviews, identify trade-offs across technical boundaries, and communicate with clarity across disciplines. It is supported by shared digital workspaces, live modeling tools, and collaborative decision platforms - but it depends fundamentally on a mindset of systems thinking and operational empathy. Integration is both technical and relational.

Behavioral Agility and Decision Influence

As the pace of operations accelerates and decision-making becomes increasingly decentralized, professionals are required to lead through influence rather than authority. Behavioral agility refers to the ability to navigate complexity, remain adaptive in dynamic environments, and maintain situational awareness under operational pressure.

This domain includes structured communication, stakeholder alignment, and framing of technical decisions for varied audiences. Influence is not about control - it is about clarity, timing, and credibility. The integration of digital communication tools and real-time collaboration platforms further elevates the need for behavioral precision and responsiveness. In this context, the most effective professionals are those who combine technical credibility with interpersonal dexterity to align teams and move decisions forward.

Designing a Capability Architecture for Evolving Technical Roles

As the demands of well construction evolve in complexity, scale, and cross-disciplinary integration, the development of technical professionals can no longer rely on fragmented training modules or passive progression through tenure. Building on the three core skill domains outlined earlier—technical fluency, cross-functional integration, and behavioral agility—this chapter introduces a structured capability architecture designed to uplift mid-career professionals into higher-impact roles. The proposed framework emphasizes deliberate development across interconnected domains, fostering the ability to synthesize knowledge, influence decisions, and leverage digital tools to accelerate delivery and reduce operational risk. Rather than focusing on generic upskilling, this architecture offers a tailored progression model that supports both individual growth and organizational resilience.

Framing Development as a Capability Journey

Capability growth is not linear. It unfolds in stages influenced by role context, exposure to complexity, and the integration of learning with operational accountability. The architecture proposed here conceptualizes professional development not as a checklist of competencies, but as a progression through increasingly integrative and outcome-oriented contributions.

Each stage of the capability journey represents a shift in mindset, scope of impact, and the ability to navigate ambiguity ([Figure 1](#)):

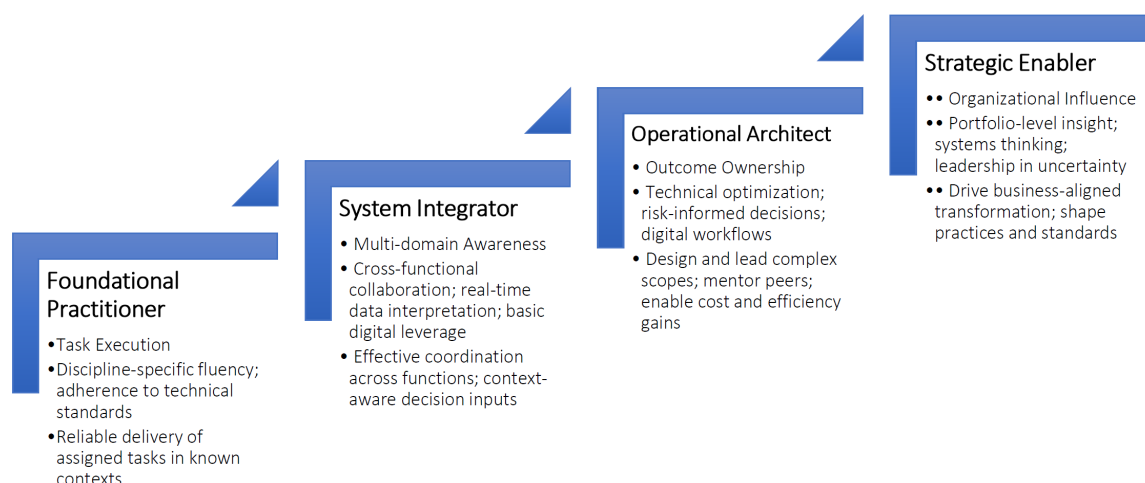


Figure 1—Capability structure

This progression is not tenure-based -it is outcome-based, observable through behavior, influence, and the complexity of problems engaged.

Interlinking the Skill Domains

The capability architecture is explicitly aligned with the three previously defined domains:

- Technical Fluency evolves from procedural execution to systems-based optimization, including the use of digital simulations, advanced diagnostics, and automation tools.
- Cross-Functional Integration expands from contextual awareness to active bridging across domains such as subsurface, HSE, operations, and commercial. Digital platforms and shared data environments become key enablers.
- Behavioral Agility and Decision Influence shifts from reactive communication to proactive leadership under uncertainty—navigating stakeholder trade-offs, influencing without authority, and managing non-technical risk.

The architecture thus serves as a developmental map, guiding individuals and organizations toward integrative capability rather than isolated skill blocks.

Multi-Dimensional Progression Assessment Framework

To move away from generic "readiness" or "observation-based" language, more structured, evidence-based progression framework, is introduced. It integrates four calibrated dimensions, each anchored in observable behaviors and measurable outcomes. This provides a transparent basis for development planning and promotion, while remaining aligned with the three core skill domains (Table 1).

Table 1—Assessment framework structure

Dimension	Description	Assessment Mechanism	Primary Domain Alignment
Technical Depth	Mastery and innovation within a technical domain	Case-based evaluation, design reviews, technical leadership examples	Technical Fluency
Situational Complexity	Ability to perform in volatile, ambiguous, or novel contexts	Complexity-weighted portfolio review, real-time simulation scoring	Cross-Functional Integration
Influence & Decision Reach	Scope of influence over cross-functional decisions and stakeholder alignment	Recorded decisions, stakeholder 360s, project outcome mapping	Behavioral Agility & Decision Influence

Dimension	Description	Assessment Mechanism	Primary Domain Alignment
Digital Leverage	Effective use of digital tools, automation, and data in engineering tasks	Digital toolchain fluency index, workflow automation initiatives	Integrated Across All Domains

Each dimension is scored on a 4-level observable scale (Initiating → Developing → Integrating → Leading), supported by real examples from the candidate's project work, peer endorsements, and leadership feedback. This structure discourages passive seniority and incentivizes outcome-driven growth.

Organizational Enablers for Scalable Capability Development

To embed and scale the capability development model across the enterprise, organizational systems must evolve from ad hoc training provision to capability ecosystem design. Four interdependent enablers ensure alignment, reinforcement, and continuity of professional growth (Figure 2):

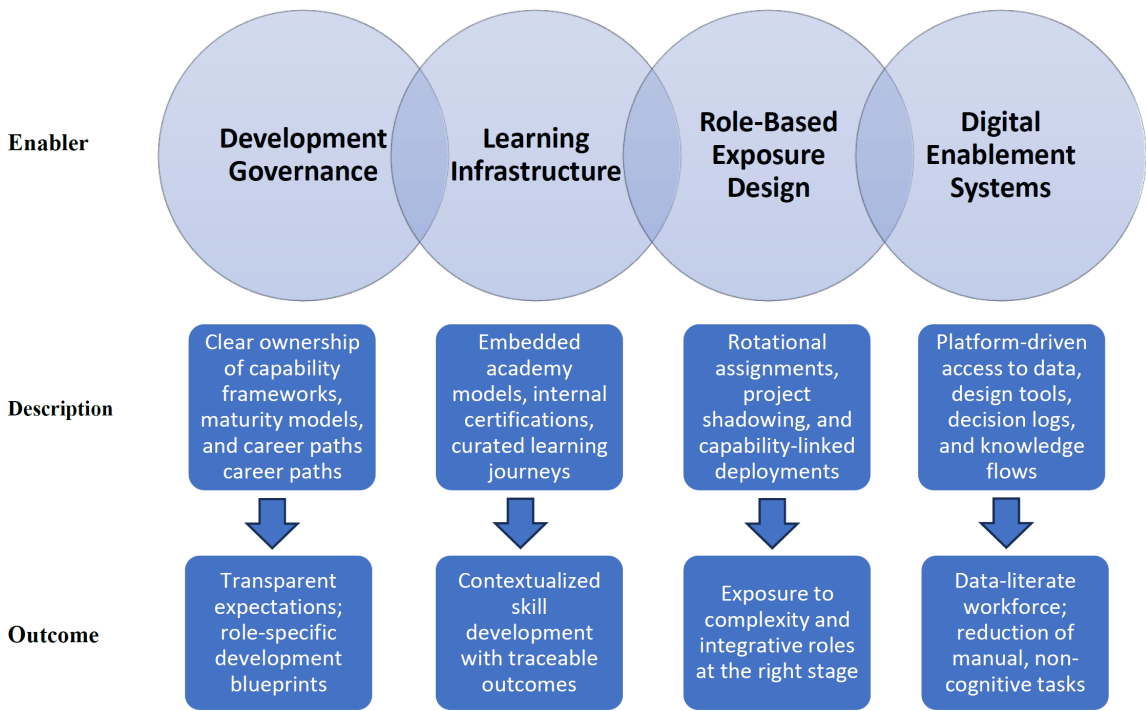


Figure 2—Organizational enablers

These systems do not merely support training; they operationalize development by creating the conditions for capability to emerge through exposure, feedback, and digital acceleration.

Implementation Considerations and Organizational Integration

Effective deployment of capability-based development depends on the seamless integration of systemic enablers across the organizational fabric. Capability progression must be reflected in how people are selected, developed, transitioned, and supported - embedding developmental logic into the daily mechanics of decision-making and resource allocation.

The architecture outlined in the previous chapter becomes actionable when it is operationalized through aligned systems, piloted in real settings, and supported by digital and behavioral feedback loops. Rather than treating capability development as a separate initiative, implementation focuses on institutionalizing capability as a design principle across key workflows and leadership routines.

The following sections detail the implementation dimensions that drive scalability, learning reinforcement, and alignment between individual growth and organizational agility.

Embedding Capability Frameworks into Organizational Systems

To avoid the common pitfall of theoretical frameworks being decoupled from day-to-day practice, capability development must be embedded within existing systems of performance management, workforce planning, and operational excellence. Integration is most effective when it occurs at three organizational levels:

- Strategic level: Align the capability model with long-term workforce planning, digital transformation strategy, and leadership succession.
- Operational level: Integrate capability assessments into technical assurance processes, project onboarding, and peer reviews.
- Tactical level: Use the capability progression matrix during individual performance reviews, learning journeys, and role transition planning.

This multi-tiered alignment ensures that development is not episodic or aspirational—it becomes part of the fabric of decision-making, resource allocation, and technical assurance.

Enabling the Transition Through Capability Pilots

Implementation at scale should be preceded by capability pilots - targeted deployments of the framework within specific business units or project teams. These pilots serve multiple purposes:

- Validate the practical utility of the framework under real-world conditions.
- Generate examples of observable behavior across progression levels to calibrate assessments.
- Identify the pain points in integration, such as system interoperability or resistance to non-seniority-based evaluation.

For example, a pilot in a high-intensity drilling asset could evaluate how mid-career engineers navigate situational complexity and influence cross-functional decisions. This not only grounds the model but also generates internal advocates who shape its adoption.

Capability flow through the Talent Lifecycle

To create institutional value, the capability framework must extend beyond individual growth and permeate core processes across the talent lifecycle. This ensures that capability is not a parallel track to existing HR systems, but rather a fundamental lens through which decisions are made — from hiring to succession.

The use cases below represent strategic integration points, where domain maturity and progression markers can meaningfully inform people decisions, reduce subjectivity, and unlock underutilized potential (Figure 3).

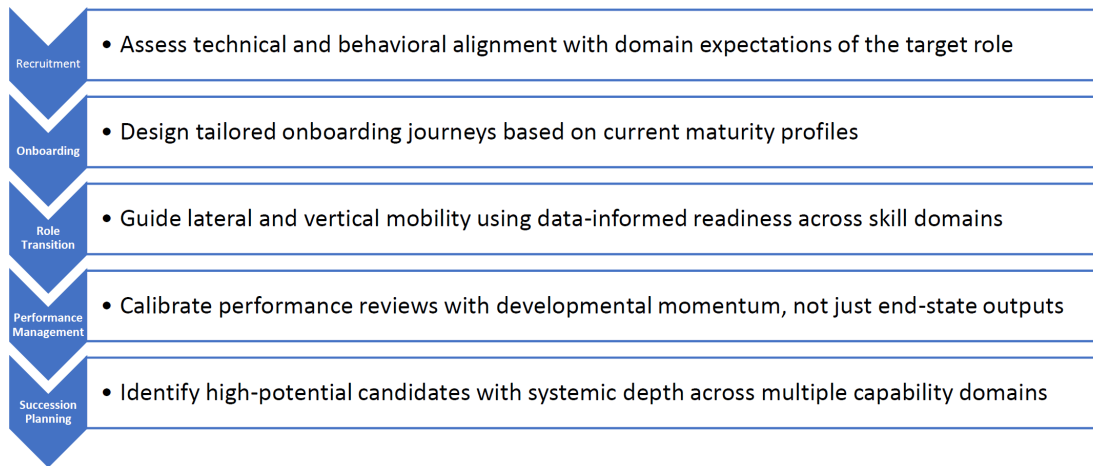


Figure 3—Lifecycle stages and related Capability application

The underlying message is a shift from role-centric thinking to capability-based orchestration - where growth is viewed as directional and developmental, not just positional or time-bound.

Digital Enablement and Feedback Infrastructure

For the proposed framework its long-term viability depends on real-time feedback, visibility, and adaptation - which are only feasible through digital enablement. Capability progression is not a static process; it is dynamic, contextual, and frequently nonlinear. As such, technology plays a vital role in converting frameworks into adaptive ecosystems.

Key elements of the digital layer include:

- Development Dashboards
- Provide visual insights into progression across domains, enabling individuals and managers to track development momentum and detect stagnation early.
- Experience Recommender Engines
- Suggest next developmental experiences (projects, challenges, or rotations) based on observed patterns of growth, role requirements, and behavioral indicators.
- Integrated Memory Systems
- Link field learnings, project outcomes, and peer feedback directly to capability elements, reinforcing continuous organizational learning.
- Behavioral Signal Capture
- Leverage digital collaboration platforms (e.g., task systems, 360s, coaching logs) to detect patterns that correspond with behavioral agility and integration markers.

These tools form a feedback loop that bridges development, deployment, and reapplication — a system where learning reinforces itself through visibility, relevance, and timely reinforcement.

Managing the Transition: Organizational Readiness and Enabling Conditions

Shifting from traditional role-based development to capability-centric growth is not a linear change — it requires organizational readiness across several dimensions. This section outlines key enablers to support successful adoption, framed constructively and with a focus on system alignment rather than cultural deficit (Table 2).

Table 2—Organizational transition dimensions and related readiness considerations

Leadership Sponsorship	Governance and Ownership	Psychological Safety	Consistency and Language	Narrative Framing
Leaders must visibly model capability-based thinking in succession and project assignments	Clear role for a Capability Office or function that connects technical, HR, and operations	Transparent development feedback requires trust and a culture of learning, not judgement	Consistent application of domain language and progression indicators across all functions	Communicate capability growth as opportunity-rich and inclusive, not evaluative or exclusive

Rather than positioning the shift as a corrective measure, the emphasis is on enabling shared accountability for development and framing capability systems as empowering — both for individuals and for the organization's strategic agility.

Institutionalizing a capability development system at scale requires synchronized action across multiple organizational layers - strategic, operational, behavioral, and digital. The following table consolidates the critical dimensions covered throughout the chapter, linking each with its implementation focus, key enablers, and potential risks. This structure is intended to support both strategic design and practical execution, allowing stakeholders to identify gaps, plan interventions, and align efforts across functions (Table 3).

Table 3—Scalable Implementation Dimensions

Dimension	Implementation Focus	Key Enablers
Operational Integration	Embed capability checkpoints in technical reviews, planning cycles, and project allocation	Line management alignment, assurance routines
Pilot Activation	Launch capability pilots targeting transition points and diverse operating contexts	Senior sponsorship, co-creation with pilot teams
Talent Lifecycle Fit	Apply domain and maturity data in hiring, onboarding, performance, and succession workflows	HR integration, calibrated scoring frameworks
Digital Infrastructure	Support feedback, visibility, and learning navigation via digital platforms	Dashboards, recommender engines, feedback systems
Organizational Readiness	Prepare functions, leadership, and teams for mindset shift and systemic adoption	Framing, governance roles, safe development environments

Conclusion

The future of well construction will not be shaped solely by new tools or digital platforms - it will be shaped by the capabilities of those who deploy, adapt, and orchestrate them. The proposed framework introduces a shift from conventional skill taxonomies to a dynamic, multidimensional capability architecture that reflects the increasing complexity, interdependence, and innovation demands of the upstream energy sector.

By redefining professional development through three core domains—Technical Fluency, Cross-Functional Integration, and Behavioral Agility—the model offers a more precise representation of modern technical expertise and leadership potential. It moves beyond time-bound or tenure-based models and provides a directional and observable progression system grounded in system behavior and cross-domain contribution.

This capability model is structured for enterprise-wide deployment, with direct application across talent systems, learning infrastructure, operational workflows, and digital enablers. Importantly, it presents a scalable and role-agnostic pathway that balances technical depth with behavioral range and integrative capacity.

While challenges such as cultural resistance, misaligned incentives, and limited feedback loops persist, they are not insurmountable. Through targeted pilots, strategic alignment, and redefined indicators of success, organizations can transition to a more adaptive and performance-oriented development model.

In summary, transitioning from static roles to dynamic capability systems marks a critical evolution for high-performance organizations. For organizations seeking to lead in performance, sustainability, and innovation, investing in capability development is not merely a workforce initiative - it is a strategic imperative for the well construction sector.

Reference

1. Abdel Fattah, D., Botros, J., Gaber, N., Boland, C. The Potential Skilling, Upskilling, and Reskilling Opportunities for the Migration and Mobility of Workers, with a Specific Focus on Gender Aspects of Workers in the Countries of Origin. *Global Strategy for Skills, Migration and Development (GS4S)*. 2024.
2. Arief Prasetyo et al Enhancing the Well Engineering Management System (WEMS) Through a Capability Maturity Model Integration (CMMI) - Based Approach. SPE-221827-MS. *ADIPEC*, Abu Dhabi, UAE, November 2024.
3. Alexey Ruzhnikov. *Maximizing Employee Potential Through Continuous Development in Technical Teams*. SPE-224754-MS. *GOTECH*, Dubai City, UAE, April 2025.
4. Brad Cawn. *Reskilling the Utility Workforce*. Quanta Technology. 2020.
5. Ellingrud, Kweilin; Gupta, Rahul; Salguero, Julian. *Building the Vital Skills for the Future of Work in Operations*. McKinsey & Company, August 2020.
6. Cecilia Malagon Uribe et al Project Management: Team Motivation, the Eye of the Storm. SPE-193039-MS. Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, November 2018.
7. EY. *The Future of Work: The Changing Skills Landscape for Miners*. Ernst & Young, Australia. 2019.
8. Irwin Kirsch, Anita Sands, Steve Robbins, Madeline Goodman and Rick Tannenbaum. *Buttressing the Middle: A Case for Reskilling and Upskilling America's Middle-Skill Workers in the 21st Century*. *The ETS center for research on human capital and education*. 2021.
9. Janet Frances Rafner, Dominik Dellermann, Dora Veraszto, Constance Elizabeth Kampf *Deskilling, Upskilling, and Reskilling: a Case for Hybrid Intelligence Article in Morals & Machines*. January 2021
10. Kehinde Oladapo Oladele, Eniola Lisoyi, Isaac I Abe. *Reskilling and Upskilling to Develop Global Relevance in the Fourth Industrial Revolution*. *Future of Work, Work-Family Satisfaction, and Employee Well-Being in the Fourth Industrial Revolution*. January 2021.
11. Marcela Escobari, Ian Seyal, Michael Meaney. *Realism About Reskilling*. *Brookings*, 2021.
12. *Rapid assessment of reskilling and upskilling needs arising from the effects of the COVID-19: Baseline study for Cameroon*. International Labour Organization. 2021.
13. World Economic Forum. *Towards a Reskilling Revolution*. Geneva. World Economic Forum. 2018.